

Chapter 4: Sturm–Liouville Theory

Mathematical Physics

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Outline

- 1 Review of linear algebra
- 2 Sturm–Liouville operators
- 3 Green's function on an interval

Where we are going

This chapter builds the operator picture behind the second-order equations of later chapters, using only basic calculus and inner products of functions.

- Functions act as vectors in an inner-product space; Fourier coefficients are extracted by orthogonality.
- A Sturm–Liouville eigenvalue problem is written, with this chapter's sign convention, as

$$\frac{d}{dx} \left(p(x) \frac{du}{dx} \right) + q(x)u(x) = -\lambda w(x)u(x),$$

where λ is the eigenvalue and typically $p(x) > 0$ and $w(x) > 0$ in the interior of the interval.

- Integration by parts identifies the *formal adjoint*. Boundary conditions that kill the boundary form give the symmetry used below.
- Green's identity then yields real eigenvalues and weighted orthogonality of eigenfunctions.
- Legendre and Bessel (drum) modes illustrate the theory. A real second-order equation can be put into divergence form by an integrating factor on an interval where its leading coefficient is nonzero. Optionally, Gram–Schmidt recovers P_0, P_1, P_2 .
- A point source then defines the Green function.

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Vector spaces and bases

A *vector space* is a collection of objects that is closed under addition and scalar multiplication. Thus, if u and v belong to the space and a, b are scalars, then

$$au + bv$$

also belongs to the space. The vectors may be functions rather than geometric arrows.

Examples.

- The polynomials of degree at most two, $a_0 + a_1x + a_2x^2$, form a three-dimensional vector space.
- The smooth functions on $[0, 1]$ satisfying $u(0) = u(1) = 0$ form a vector space, because

$$(au + bv)(0) = au(0) + bv(0) = 0, \quad (au + bv)(1) = au(1) + bv(1) = 0.$$

Vector spaces and bases (cont.)

A *basis* is a set of vectors from which every vector in the space has a unique linear combination. Existence is called *spanning*; uniqueness follows from *linear independence* (only zero coefficients give the zero vector). For the quadratic polynomials, $\{1, x, x^2\}$ is a basis, and

$$1 + 2x + 3x^2 = 1 \cdot 1 + 2 \cdot x + 3 \cdot x^2. \quad (4.1.1)$$

The set

$$\{1, 2x - 1, 6x^2 - 6x + 1\}$$

is another basis. The different degrees make the three polynomials independent: matching the highest power first forces each coefficient of a zero combination to vanish. To express the same function in this basis, write

$$\begin{aligned} 1 + 2x + 3x^2 &= A + B(2x - 1) + C(6x^2 - 6x + 1) \\ &= (A - B + C) + (2B - 6C)x + 6Cx^2. \end{aligned}$$

Vector spaces and bases (cont.)

Equating coefficients of $1, x, x^2$ gives

$$6C = 3, \quad C = \frac{1}{2},$$

$$2B - 6C = 2, \quad 2B - 3 = 2, \quad B = \frac{5}{2},$$

$$A - B + C = 1, \quad A - \frac{5}{2} + \frac{1}{2} = 1, \quad A = 3.$$

Therefore

$$1 + 2x + 3x^2 = 3 \cdot 1 + \frac{5}{2}(2x - 1) + \frac{1}{2}(6x^2 - 6x + 1). \quad (4.1.2)$$

The number of elements in a finite basis is the *dimension*. To discuss infinite bases, we first need a notion of distance and convergence.

Inner products, norms, and coefficients

Step 1: choose the inner product. We use the convention that the inner product is *linear in its first slot*. An inner product $(\cdot, \cdot)$ satisfies

- ❶ $(au + bv, w) = a(u, w) + b(v, w),$
- ❷ $(u, v) = \overline{(v, u)},$
- ❸ $(u, u) \geq 0,$ with equality only for the zero vector.

In particular, for a scalar $c,$

$$(cu, v) = c(u, v). \quad (4.1.3)$$

Conjugate symmetry forces *conjugate-linearity in the second slot*:

$$(u, cv) = \overline{c}(u, v). \quad (4.1.4)$$

For example, with $u = (1, 0)$ and $(u, v) = u_1 \overline{v_1} + u_2 \overline{v_2},$

$$(iu, u) = i, \quad (u, iu) = -i.$$

The conjugate is essential: $(iu, iu) = 1,$ not $-1.$

The standard L^2 inner product on an interval $[a, b]$ is

$$(u, v) = \int_a^b u(x) \overline{v(x)} dx. \quad (4.1.5)$$

Inner products, norms, and coefficients (cont.)

If w is a real weight with $w(x) > 0$ on the interior (it may vanish at isolated endpoints), the weighted inner product is

$$(u, v)_w = \int_a^b u(x) \overline{v(x)} w(x) dx. \quad (4.1.6)$$

For example, the Bessel radial weight on $[0, 1]$ is $w(s) = s$. Then

$$(u, u)_w = \int_a^b |u(x)|^2 w(x) dx$$

is positive for every nonzero continuous u of interest in this course.

The norm induced by an inner product is

$$\|u\| = \sqrt{(u, u)}, \quad \|u\|_w = \sqrt{(u, u)_w}. \quad (4.1.7)$$

A nonzero function is normalized by replacing it with $u/\|u\|$.

Two vectors are *orthogonal* when $(u, v) = 0$. An orthonormal set $\{u_i\}$ satisfies

$$(u_i, u_j) = \delta_{ij}, \quad (4.1.8)$$

where $\delta_{ij} = 1$ for $i = j$ and 0 otherwise.

Inner products, norms, and coefficients (cont.)

Step 2: extract coefficients by orthogonality. Suppose first that

$$f = \sum_{i=1}^N a_i u_i$$

in an orthonormal basis. Taking the inner product with u_j gives

$$\begin{aligned}(f, u_j) &= \left(\sum_{i=1}^N a_i u_i, u_j \right) \\ &= \sum_{i=1}^N a_i (u_i, u_j) \\ &= \sum_{i=1}^N a_i \delta_{ij} \\ &= a_j.\end{aligned}\tag{4.1.9}$$

(The same formula holds for an infinite expansion once one knows the series converges in the L^2 norm, so the inner product may pass under the limit.)

Inner products, norms, and coefficients (cont.)

If the basis is orthogonal but not normalized, then

$$(f, u_j) = \sum_i a_i (u_i, u_j) = a_j (u_j, u_j) = a_j \|u_j\|^2, \quad (4.1.10)$$

so

$$a_j = \frac{(f, u_j)}{\|u_j\|^2}. \quad (4.1.11)$$

Infinite-dimensional bases and L^2

For an infinite series of functions, we must say how its partial sums approach the function. In $L^2(0, 1)$ the distance is measured by the square integral:

$$\|f\|_2 = \left(\int_0^1 |f(x)|^2 dx \right)^{1/2} < \infty.$$

Functions differing only at finitely many points, or on a set that does not affect integrals, represent the same L^2 vector. No measure theory is needed here.

Use the sine modes

$$u_n(x) = \sin(n\pi x), \quad n = 1, 2, \dots \quad (4.1.12)$$

The next calculation shows that they are orthogonal, with $\|u_n\|^2 = 1/2$. Orthogonality alone is not enough: removing u_1 leaves an orthogonal set that cannot reproduce u_1 . The additional property is called *completeness*.

Infinite-dimensional bases and L^2 (cont.)

Completeness of the sine system (quoted). Every $f \in L^2(0, 1)$ has the expansion

$$f = \sum_{n=1}^{\infty} a_n u_n, \quad a_n = \frac{(f, u_n)}{\|u_n\|^2}, \quad (4.1.13)$$

with mean-square convergence:

$$\left\| f - \sum_{n=1}^N a_n u_n \right\|_2 \longrightarrow 0.$$

This is the infinite-series version of the projection formula (4.1.11). Its proof of completeness belongs to Fourier analysis; we will compute the norm and use the coefficients.

Mean-square convergence concerns the integrated squared error, not the value at every point. Every sine sum is zero at 0 and 1, even when f is not.

Example: orthogonality of the sine modes

Let $u_m(x) = \sin(m\pi x)$ and $u_n(x) = \sin(n\pi x)$, with m, n positive integers. Use the product-to-sum identity

$$\sin A \sin B = \frac{1}{2}[\cos(A - B) - \cos(A + B)]. \quad (4.1.14)$$

For $m \neq n$, integration gives

$$\begin{aligned} \int_0^1 u_m u_n dx &= \left[\frac{\sin((m-n)\pi x)}{2(m-n)\pi} - \frac{\sin((m+n)\pi x)}{2(m+n)\pi} \right]_0^1 \\ &= 0, \end{aligned} \quad (4.1.15)$$

because sine is zero at every integer multiple of π .

For $m = n$, use $\sin^2(m\pi x) = [1 - \cos(2m\pi x)]/2$ instead:

$$\int_0^1 u_m^2 dx = \left[\frac{x}{2} - \frac{\sin(2m\pi x)}{4m\pi} \right]_0^1 = \frac{1}{2}. \quad (4.1.16)$$

Together,

$$(u_m, u_n) = \frac{1}{2} \delta_{mn}. \quad (4.1.17)$$

Thus multiplying each u_n by $\sqrt{2}$ would give a norm-one basis.

Example: Fourier sine-series coefficients

For $f \in L^2(0, 1)$, write

$$f(x) = \sum_{n=1}^{\infty} a_n \sin(n\pi x) \quad (4.1.18)$$

in the mean-square sense just defined. The projection formula (4.1.11) and the norm $\|u_k\|^2 = 1/2$ give

$$\begin{aligned} a_k &= \frac{(f, u_k)}{\|u_k\|^2} = \frac{\int_0^1 f(x) \sin(k\pi x) dx}{1/2} \\ &= 2 \int_0^1 f(x) \sin(k\pi x) dx. \end{aligned} \quad (4.1.19)$$

The factor 2 comes from the norm, not from an additional convention. The same projection rule will give coefficients for Bessel and polynomial expansions; only the interval, weight, and norm will change.

Linear operators and formal adjoints

A linear operator acts on functions. It satisfies $L(au + bv) = aLu + bLv$. Differentiation, multiplication by a fixed function, and $Au(x) = \int_{x_0}^x u(t) dt$ are examples. The allowed functions must also satisfy any specified boundary conditions.

Integration by parts moves a derivative. With $D = d/dx$,

$$\begin{aligned}(u, Dv) &= \int_0^1 u \overline{v'} dx \\ &= [u\overline{v}]_0^1 - \int_0^1 u' \overline{v} dx = [u\overline{v}]_0^1 + (-Du, v).\end{aligned}\tag{4.1.20}$$

The expression left behind is the *formal adjoint*: $D^\dagger = -D$. Even if the boundary term vanishes, a minus sign remains. Thus D is not formally self-adjoint.

Linear operators and formal adjoints (cont.)

Two integrations restore the sign. For $L = D^2$,

$$\begin{aligned}(u, Lv) &= \int_0^1 u \overline{v''} \, dx \\ &= [u \overline{v'}]_0^1 - \int_0^1 u' \overline{v'} \, dx\end{aligned}\tag{4.1.21}$$

$$= [u \overline{v'} - u' \overline{v}]_0^1 + \int_0^1 u'' \overline{v} \, dx.\tag{4.1.22}$$

Hence $L^\dagger = L$. If $u(0) = u(1) = v(0) = v(1) = 0$, both boundary contributions vanish:

$$(u, Lv) = (Lu, v).\tag{4.1.23}$$

This symmetry is the property that makes eigenfunctions with different eigenvalues orthogonal.

Both the expression and the boundary conditions matter. On a regular interval, homogeneous Dirichlet (zero value), Neumann (zero derivative), or real Robin (zero linear combination) conditions make this expression a self-adjoint problem. Singular endpoints require a separate limiting boundary check.

Eigenfunctions and the orthogonality theorem

An eigenpair (u, λ) of an operator L satisfies

$$Lu = \lambda u, \quad u \neq 0. \quad (4.1.24)$$

A generalized eigenvalue problem introduces a prescribed real weight $w(x)$:

$$Lu = \lambda wu. \quad (4.1.25)$$

In the Sturm–Liouville convention used later, the right-hand side is $-\lambda wu$.

Examples.

- On a domain containing $e^{\lambda x}$ without incompatible boundary conditions,

$$\frac{d}{dx} e^{\lambda x} = \lambda e^{\lambda x}.$$

- For real $m \neq 0$, since

$$\frac{d^2}{dx^2} \cos(mx) = -m^2 \cos(mx), \quad \frac{d^2}{dx^2} \sin(mx) = -m^2 \sin(mx),$$

both functions are eigenfunctions of the second-derivative differential expression, with eigenvalue $-m^2$. At $m = 0$, $\sin(0x)$ is the zero function and therefore is not an eigenfunction; the nonzero constant $\cos(0x) = 1$ is.

Eigenfunctions and the orthogonality theorem (cont.)

Theorem

Every eigenvalue of a self-adjoint operator is real. Eigenfunctions belonging to distinct eigenvalues are orthogonal.

We will prove the weighted version in §4.2 by subtracting the two differential equations and checking their boundary terms. The essential assumption is that both eigenfunctions satisfy the same self-adjoint boundary conditions.

Distinct eigenvalues are required for orthogonality. For a repeated eigenvalue, orthogonal linear combinations can be chosen within its solution space; the Gram–Schmidt calculation at the end illustrates this procedure.

Eigenfunctions and the orthogonality theorem (cont.)

Application: the fixed string. Let $Lu = u''$ act on twice continuously differentiable functions on $[0, 1]$ that satisfy the Dirichlet conditions $u(0) = u(1) = 0$. First, every eigenvalue is negative. If $Lu = \mu u$ with $u \neq 0$, then

$$\begin{aligned}\mu \|u\|^2 &= (Lu, u) = \int_0^1 u'' \bar{u} \, dx \\ &= [u' \bar{u}]_0^1 - \int_0^1 |u'|^2 \, dx \\ &= - \int_0^1 |u'|^2 \, dx < 0.\end{aligned}$$

(The endpoint term vanishes because $u(0) = u(1) = 0$. The strict inequality holds because $u' = 0$ would make u constant, and the two Dirichlet conditions would then force $u = 0$.) Hence write $\mu = -k^2$ with $k > 0$. The eigenvalue equation becomes

$$u'' + k^2 u = 0, \quad u(0) = u(1) = 0.$$

Its general solution is

$$u(x) = A \cos(kx) + B \sin(kx).$$

Eigenfunctions and the orthogonality theorem (cont.)

The first boundary condition gives

$$0 = u(0) = A \cos 0 + B \sin 0 = A,$$

so $A = 0$. The second gives

$$0 = u(1) = B \sin k.$$

A nonzero eigenfunction requires $B \neq 0$, hence

$$\sin k = 0 \implies k = n\pi, \quad n = 1, 2, 3, \dots$$

Thus

$$u_n(x) = \sin(n\pi x), \quad Lu_n = -(n\pi)^2 u_n. \quad (4.1.26)$$

Because $(u, Lv) = (Lu, v)$ holds under these boundary conditions, the theorem gives $(u_m, u_n) = 0$ for $m \neq n$, in agreement with the direct integral calculation in (4.1.17).

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Formal adjoint of a second-order expression

For real, sufficiently differentiable coefficients, write

$$\ell[u] = p_0 u'' + p_1 u' + p_2 u. \quad (4.2.1)$$

Integrating by parts moves every derivative off u and onto the coefficient–function product with v : the $p_0 u''$ term is integrated twice (the D^2 rule), returning $(p_0 v)''$; the $p_1 u'$ term once (the D rule, with its minus sign), returning $-(p_1 v)'$; and $p_2 u$ is unchanged. The coefficients stay inside the derivatives because they multiply v . Hence

$$\ell^\dagger[v] = (p_0 v)'' - (p_1 v)' + p_2 v. \quad (4.2.2)$$

The coefficient of v' on the right is $2p_0' - p_1$. For it to equal the coefficient p_1 in $\ell[v]$, we need $p_1 = p_0'$. This also makes the coefficients of v agree, so

$$\ell^\dagger = \ell \iff p_1 = p_0'. \quad (4.2.3)$$

The practical test is the product rule:

$$p_0 u'' + p_0' u' = (p_0 u')'.$$

We next derive the boundary identity for this form. Formal symmetry by itself does not specify which boundary conditions are allowed.

The Sturm–Liouville expression and Green's identity

Step 1: put the expression in divergence form. When $p_1 = p_0'$, the first two terms in (4.2.1) combine by the product rule:

$$\begin{aligned}(p_0 u')' &= p_0' u' + p_0 u'' \\ &= p_1 u' + p_0 u''.\end{aligned}$$

Writing $p = p_0$ and $q = p_2$ gives the Sturm–Liouville differential expression

$$\ell[u] = (pu')' + qu. \quad (4.2.4)$$

Step 2: derive Green's identity. Because p and q are real and complex conjugation commutes with differentiation, $\overline{\ell[v]} = \overline{(pv')' + qv} = (p\overline{v'})' + q\overline{v}$. Integrating by parts once in each of the following expressions,

$$\begin{aligned}(u, \ell[v]) &= \int_a^b u [(p\overline{v'})' + q\overline{v}] dx \\ &= [up\overline{v'}]_a^b - \int_a^b pu'\overline{v'} dx + \int_a^b qu\overline{v} dx,\end{aligned} \quad (4.2.5)$$

The Sturm–Liouville expression and Green's identity (cont.)

and

$$\begin{aligned}(\ell[u], v) &= \int_a^b [(pu')' + qu] \bar{v} \, dx \\ &= [pu'\bar{v}]_a^b - \int_a^b pu'\bar{v}' \, dx + \int_a^b qu\bar{v} \, dx.\end{aligned}\tag{4.2.6}$$

Subtracting cancels both integrals and yields Green's identity, also called Lagrange's identity:

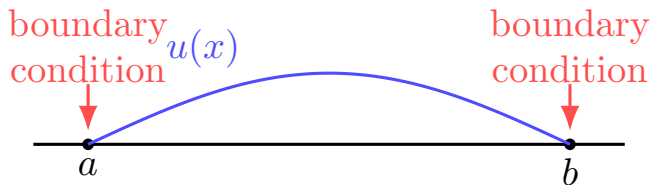
$$(u, \ell[v]) - (\ell[u], v) = [p(u\bar{v}' - u'\bar{v})]_a^b.\tag{4.2.7}$$

The endpoint expression

$$\mathcal{B}[u, v](x) = p(x) (u(x)\overline{v'(x)} - u'(x)\overline{v(x)})\tag{4.2.8}$$

is the Lagrange boundary form.

The Sturm–Liouville expression and Green's identity (cont.)



$$p(u\overline{v}' - u'\overline{v})\Big|_a^b$$

self-adjoint boundary conditions make
the endpoint contribution zero

The Sturm–Liouville expression and Green's identity (cont.)

Step 3: choose boundary conditions that kill the boundary form. For a regular finite interval, standard self-adjoint boundary conditions include the following.

Separated real Robin conditions. At each endpoint impose

$$\alpha_a u(a) + \beta_a (pu')(a) = 0, \quad \alpha_b u(b) + \beta_b (pu')(b) = 0, \quad (4.2.9)$$

where the coefficients are real and $(\alpha, \beta) \neq (0, 0)$ at each endpoint. The same conditions are imposed on v .

To verify cancellation at one endpoint, suppose first that $\beta \neq 0$. Then

$$pu' = -\frac{\alpha}{\beta} u, \quad p\overline{v'} = -\frac{\alpha}{\beta} \overline{v},$$

so

$$\begin{aligned} p(u\overline{v'} - u'\overline{v}) &= u(p\overline{v'}) - (pu')\overline{v} \\ &= u\left(-\frac{\alpha}{\beta}\overline{v}\right) - \left(-\frac{\alpha}{\beta}u\right)\overline{v} \\ &= 0. \end{aligned}$$

The Sturm–Liouville expression and Green’s identity (cont.)

If $\beta = 0$, then $u = v = 0$ at that endpoint (Dirichlet). If $\alpha = 0$, then $pu' = pv' = 0$ there (Neumann or natural boundary condition). Thus (4.2.9) contains Dirichlet and Neumann conditions as special cases.

Periodic conditions. The coupled conditions

$$u(a) = u(b), \quad (pu')(a) = (pu')(b) \quad (4.2.10)$$

and the same conditions for v make $\mathcal{B}[u, v](b) = \mathcal{B}[u, v](a)$, so the difference in (4.2.7) is zero.

Singular endpoints. If p vanishes at an endpoint or the interval is infinite, the bare statement “ $p = 0$ ” is not enough by itself. One checks the limiting boundary form and keeps only those modes for which

$$\lim_{x \rightarrow a^+} \mathcal{B}[u, v](x) = 0, \quad \lim_{x \rightarrow b^-} \mathcal{B}[u, v](x) = 0. \quad (4.2.11)$$

For Legendre polynomials, $p = 1 - x^2$ tends to zero while the polynomials and their derivatives stay finite. For Hermite polynomials, $p = e^{-x^2}$ kills every polynomial Wronskian as $x \rightarrow \pm\infty$: the modes are real, so $u\overline{v}' - u'\overline{v} = uv' - u'v = W[u, v]$, and $\mathcal{B} = e^{-x^2} W[u, v] \rightarrow 0$ because the Gaussian decays faster than any polynomial grows.

The Sturm–Liouville expression and Green's identity (cont.)

When these boundary conditions or verified singular limits make the boundary form vanish, $(u, \ell[v]) = (\ell[u], v)$. The next argument also requires finite weighted squared norms; it establishes reality and orthogonality, not a general self-adjointness or completeness theorem.

The Sturm–Liouville eigenvalue problem

Step 1: write the two eigenvalue equations. With the sign convention of this chapter, the Sturm–Liouville eigenvalue equation is

$$(pu')' + qu = -\lambda wu, \quad (4.2.12)$$

Here p is continuously differentiable and q, w are continuous in the interior; all three are real, with $p > 0$ and $w > 0$ there. We use nonzero eigenfunctions with finite weighted squared norm, $0 < \int_a^b w|u|^2 dx < \infty$. At a singular endpoint, take limits from inside the interval. The chosen boundary conditions must make the limiting Lagrange boundary form vanish.

Let u_i and u_j satisfy

$$(pu'_i)' + qu_i = -\lambda_i wu_i, \quad (4.2.13)$$

$$(pu'_j)' + qu_j = -\lambda_j wu_j. \quad (4.2.14)$$

Conjugating (4.2.14), using reality of p, q, w , gives

$$(p\overline{u'_j})' + q\overline{u_j} = -\overline{\lambda_j} w\overline{u_j}. \quad (4.2.15)$$

The Sturm–Liouville eigenvalue problem (cont.)

Multiply (4.2.13) by $\overline{u_j}$ and (4.2.15) by u_i :

$$\begin{aligned}\overline{u_j}(pu_i')' + qu_i\overline{u_j} &= -\lambda_i wu_i\overline{u_j}, \\ u_i(p\overline{u_j}')' + qu_i\overline{u_j} &= -\overline{\lambda_j} wu_i\overline{u_j}.\end{aligned}$$

Subtracting the second equation from the first cancels the q terms:

$$\overline{u_j}(pu_i')' - u_i(p\overline{u_j}')' = -(\lambda_i - \overline{\lambda_j})wu_i\overline{u_j}. \quad (4.2.16)$$

Step 2: turn the difference into a boundary term. The left-hand side is an exact derivative. Indeed,

$$\begin{aligned}\frac{d}{dx} [p(\overline{u_j}u_i' - u_i\overline{u_j}')] &= \overline{u_j}(pu_i')' + p\overline{u_j}'u_i' - u_i(p\overline{u_j}')' - pu_i'\overline{u_j}' \\ &= \overline{u_j}(pu_i')' - u_i(p\overline{u_j}')' .\end{aligned}$$

Thus (4.2.16) is

$$\frac{d}{dx} [p(\overline{u_j}u_i' - u_i\overline{u_j}')] = -(\lambda_i - \overline{\lambda_j})wu_i\overline{u_j}. \quad (4.2.17)$$

The Sturm–Liouville eigenvalue problem (cont.)

Integrating from a to b gives

$$\left[p(\overline{u_j} u_i' - u_i \overline{u_j}') \right]_a^b = -(\lambda_i - \overline{\lambda_j}) \int_a^b w u_i \overline{u_j} dx. \quad (4.2.18)$$

The integrand in the bracket is $p(\overline{u_j} u_i' - u_i \overline{u_j}') = -\mathcal{B}[u_i, u_j](x)$, the negative of the Lagrange boundary form (4.2.8). Thus the separated Robin conditions (4.2.9), periodic conditions (4.2.10), or verified singular limits (4.2.11) make the left-hand side zero.

Step 3: conclude reality and weighted orthogonality. Set $j = i$. Since $u_i \neq 0$ and $w > 0$ on the interior,

$$0 < \int_a^b w |u_i|^2 dx < \infty.$$

Equation (4.2.18) therefore gives

$$\lambda_i - \overline{\lambda_i} = 0,$$

The Sturm–Liouville eigenvalue problem (cont.)

so every eigenvalue is real. For two distinct eigenvalues, reality gives $\overline{\lambda_j} = \lambda_j$; substituting this into (4.2.18) and using that the boundary conditions make the left-hand bracket vanish leaves

$$0 = -(\lambda_i - \lambda_j) \int_a^b w u_i \overline{u_j} dx,$$
$$\lambda_i \neq \lambda_j \implies \int_a^b w u_i \overline{u_j} dx = 0.$$

Hence

$$(u_i, u_j)_w = 0 \quad (\lambda_i \neq \lambda_j). \quad (4.2.19)$$

Equivalently, $A = -w^{-1}\ell$ satisfies $(Au, v)_w = (u, Av)_w$ under these boundary conditions.

This is the weighted symmetry used in the calculation.

Example: Legendre equation

For a nonnegative integer n , the Legendre equation is

$$(1 - x^2)y'' - 2xy' + n(n + 1)y = 0, \quad -1 < x < 1. \quad (4.2.20)$$

By the product rule,

$$\begin{aligned} \frac{d}{dx} [(1 - x^2)y'] &= (1 - x^2)y'' + (1 - x^2)'y' \\ &= (1 - x^2)y'' - 2xy'. \end{aligned}$$

Therefore (4.2.20) is equivalent to

$$\frac{d}{dx} [(1 - x^2)y'] = -n(n + 1)y. \quad (4.2.21)$$

The Sturm–Liouville data are

$$p(x) = 1 - x^2, \quad q(x) = 0, \quad w(x) = 1, \quad \lambda_n = n(n + 1).$$

The bounded solution is the Legendre polynomial P_n , normalized by $P_n(1) = 1$. The first examples are

$$P_0 = 1, \quad P_1 = x, \quad P_2 = \frac{1}{2}(3x^2 - 1).$$

Example: Legendre equation (cont.)

They can be checked directly in the equation. Chapter 6 constructs the whole family by series and Rodrigues' formula. Here we use their polynomial nature to examine the endpoint term: both P_n and P'_n stay finite at $x = \pm 1$.

For two degrees $n \neq k$, the eigenvalues $n(n+1)$ and $k(k+1)$ differ. The endpoint term is

$$(1-x^2)(P_n P'_k - P'_n P_k) \longrightarrow 0 \quad (x \rightarrow \pm 1),$$

because both polynomials and their derivatives stay finite. Therefore the weighted orthogonality theorem (4.2.19) applies directly:

$$\int_{-1}^1 P_n(x) P_k(x) dx = 0 \quad (n \neq k). \quad (4.2.22)$$

Thus the Legendre polynomials are orthogonal with weight $w(x) = 1$.

Example: Bessel equation and circular-drum modes

For a fixed nonnegative integer angular order m , the Bessel function $J_m(x)$ satisfies

$$x^2 J_m''(x) + x J_m'(x) + (x^2 - m^2) J_m(x) = 0. \quad (4.2.23)$$

For a radial mode on the unit disk, define

$$y(s) = J_m(\alpha s), \quad \alpha > 0, \quad 0 \leq s \leq 1.$$

The chain rule gives

$$\begin{aligned} y'(s) &= \alpha J_m'(\alpha s), \\ y''(s) &= \alpha^2 J_m''(\alpha s), \end{aligned}$$

Substituting these derivative factors into Bessel's equation at $x = \alpha s$ gives

$$s^2 y'' + sy' + (\alpha^2 s^2 - m^2) y = 0. \quad (4.2.24)$$

For $s > 0$, divide by s :

$$s y'' + y' + \left(\alpha^2 s - \frac{m^2}{s} \right) y = 0.$$

Example: Bessel equation and circular-drum modes (cont.)

Since

$$(sy')' = sy'' + y',$$

the equation becomes

$$(sy')' - \frac{m^2}{s}y + \alpha^2 sy = 0,$$

or

$$\frac{d}{ds} \left(s \frac{dy}{ds} \right) - \frac{m^2}{s}y = -\alpha^2 sy. \quad (4.2.25)$$

Thus

$$p(s) = s, \quad q(s) = -\frac{m^2}{s}, \quad w(s) = s, \quad \lambda = \alpha^2.$$

The endpoint $s = 0$ is singular, so the equation divided by s is interpreted on $(0, 1]$ together with regularity at the center.

For fixed m , let

$$y_i(s) = J_m(\alpha_i s), \quad y_j(s) = J_m(\alpha_j s).$$

Their eigenvalues are α_i^2 and α_j^2 . Applying (4.2.18) on $[\varepsilon, 1]$ with $p = w = s$ gives

$$[s(y_j y_i' - y_i y_j')]_{\varepsilon}^1 = -(\alpha_i^2 - \alpha_j^2) \int_{\varepsilon}^1 s y_i y_j ds. \quad (4.2.26)$$

Example: Bessel equation and circular-drum modes (cont.)

We must now check both endpoints; this is the part specific to the drum.

At $s = 1$, impose one common boundary condition on the whole family:

- Dirichlet: $y_i(1) = J_m(\alpha_i) = 0$ for every i ; or
- Neumann: $y_i'(1) = \alpha_i J_m'(\alpha_i) = 0$ for every i , where J_m' denotes differentiation with respect to the Bessel-function argument.

Either choice makes the boundary expression at $s = 1$ zero. For the Neumann condition with $m = 0$, $\alpha = 0$ also gives the constant mode $y(s) = 1$ with eigenvalue $\lambda = 0$; the positive α_i below label the nonconstant modes.

At $s = 0$, the regular Bessel solution has the expansions

$$J_m(z) = \frac{1}{m!} \left(\frac{z}{2}\right)^m + O(z^{m+2}) \quad (m \geq 1),$$

$$J_0(z) = 1 - \frac{z^2}{4} + O(z^4).$$

Hence, for $m \geq 1$,

$$y_i(s) = O(s^m), \quad y_i'(s) = O(s^{m-1}),$$

so

$$s(y_i y_i' - y_i' y_i) = O(s^{2m}) \longrightarrow 0.$$

Example: Bessel equation and circular-drum modes (cont.)

For $m = 0$, $y_i(s) = 1 + O(s^2)$ and $y_i'(s) = O(s)$, giving the same boundary limit, now of order $O(s^2)$. The singular Y_m solutions are excluded by regularity at the disk center.

Letting $\varepsilon \rightarrow 0^+$ in (4.2.26) and taking distinct positive α_i, α_j gives

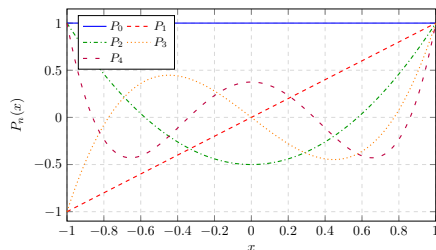
$$0 = -(\alpha_i^2 - \alpha_j^2) \int_0^1 s J_m(\alpha_i s) J_m(\alpha_j s) ds.$$

Since $\alpha_i^2 \neq \alpha_j^2$,

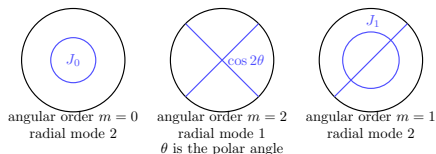
$$\int_0^1 J_m(\alpha_i s) J_m(\alpha_j s) s ds = 0 \quad (i \neq j). \quad (4.2.27)$$

The radial weight is therefore $w(s) = s$.

Legendre and Bessel modes



Legendre modes are orthogonal on $[-1, 1]$
with weight $w(x) = 1$.



Drum modes use the radial weight
 $w(s) = s$; a common boundary condition
at $s = 1$ selects the admissible α_i .

Putting a general equation into Sturm–Liouville form

Step 1: fully worked example—Hermite. Start from the concrete equation

$$y'' - 2xy' + \lambda y = 0. \quad (4.2.28)$$

Choose the integrating factor

$$Q(x) = \exp\left(\int_0^x -2t \, dt\right) = e^{-x^2}, \quad Q'(x) = -2xQ(x).$$

Multiplication of (4.2.28) by Q gives

$$Qy'' - 2xQy' + \lambda Qy = 0.$$

The first two terms combine by the product rule:

$$(Qy')' = Qy'' + Q'y' = Qy'' - 2xQy'.$$

Hence

$$(e^{-x^2} y')' = -\lambda e^{-x^2} y. \quad (4.2.29)$$

Reading off the data,

$$p(x) = e^{-x^2}, \quad q(x) = 0, \quad w(x) = e^{-x^2}.$$

Putting a general equation into Sturm–Liouville form (cont.)

The polynomial solutions occur at $\lambda_n = 2n$ and are the physicists' Hermite polynomials H_n , normalized to leading coefficient 2^n :

$$H_0 = 1, \quad H_1 = 2x, \quad H_2 = 4x^2 - 2.$$

Chapter 6 derives these by series. For example, $H_2'' - 2xH_2' + 4H_2 = 8 - 16x^2 + 16x^2 - 8 = 0$. For nonnegative integers $n \neq m$, both H_n, H_m and their derivatives are polynomials, so

$$e^{-x^2} (H_n H_m' - H_n' H_m) \longrightarrow 0 \quad (x \rightarrow \pm\infty),$$

and $\lambda_n - \lambda_m = 2(n - m) \neq 0$. Weighted orthogonality therefore yields

$$\int_{-\infty}^{\infty} H_n(x) H_m(x) e^{-x^2} dx = 0 \quad (n \neq m). \quad (4.2.30)$$

Step 2: general formula. Now abstract the calculation. Consider

$$p_0(x)u'' + p_1(x)u' + p_2(x)u = 0 \quad (4.2.31)$$

Putting a general equation into Sturm–Liouville form (cont.)

with real coefficients on an interval where $p_0(x) \neq 0$ and p_1/p_0 is continuous. Divide by p_0 :

$$u'' + \frac{p_1}{p_0} u' + \frac{p_2}{p_0} u = 0. \quad (4.2.32)$$

Define the integrating factor

$$Q(x) = \exp \left(\int_{x_0}^x \frac{p_1(t)}{p_0(t)} dt \right). \quad (4.2.33)$$

The chain rule gives $Q'/Q = p_1/p_0$. Multiplying (4.2.32) by Q ,

$$Qu'' + Q' u' + Q \frac{p_2}{p_0} u = 0.$$

The first two terms combine to $(Qu')'$, so

$$(Qu')' + Q \frac{p_2}{p_0} u = 0. \quad (4.2.34)$$

Putting a general equation into Sturm–Liouville form (cont.)

Thus the formally self-adjoint expression has

$$p = Q, \quad q = Q \frac{p_2}{p_0}.$$

Equivalently, without first dividing by p_0 , multiply the original equation (4.2.31) by Q/p_0 (not by Q alone).

If the spectral parameter occurs as

$$p_0 u'' + p_1 u' + (r_0(x) + \lambda r_1(x))u = 0, \quad (4.2.35)$$

the same steps give

$$(Qu')' + qu = -\lambda wu, \quad q = Q \frac{r_0}{p_0}, \quad w = Q \frac{r_1}{p_0}. \quad (4.2.36)$$

The positive-weight theory applies when r_1/p_0 has a fixed nonzero sign on the interval. If it is negative, replace λ by $-\lambda$ and w by $-w$. A sign-changing weight is not covered by that theory.

Table of contents

- 1 Review of linear algebra
- 2 Sturm–Liouville operators
- 3 Green's function on an interval

A unit point source creates a jump

The machinery above solves the homogeneous problem $Lu = 0$. A *point source* turns it into a solution formula for $Lu = f$. Take

$$L[u] = -(pu')' - qu,$$

the sign convention in which the eigenvalue problem reads $Lu = \lambda wu$, with separated self-adjoint boundary conditions on $[a, b]$.

Define the Dirac delta $\delta(x - x')$ operationally: it vanishes for $x \neq x'$ and, for continuous f ,

$$\int_a^b \delta(x - x') f(x) dx = f(x'), \quad a < x' < b.$$

It models a unit point source at x' , not an ordinary function.

The *Green function* $G(x, x')$ solves

$$L_x G = \delta(x - x')$$

under the problem's boundary conditions. Throughout this frame assume 0 is *not* an eigenvalue of L —equivalently, that $Lu = 0$ with these boundary conditions forces $u = 0$. This is what makes L invertible, so a unique Green function exists: it guarantees $W[y_L, y_R] \neq 0$ in the two-solution form below and $\lambda_n \neq 0$ in the modal series. If 0 is an eigenvalue (as for the $m = 0$ Neumann drum of §4.2), the point-source problem is

A unit point source creates a jump (cont.)

solvable only up to the null space and subject to a solvability condition, and both formulas must be modified. Away from x' the equation is homogeneous, so G is assembled from two solutions: y_L satisfying the left condition and y_R the right,

$$G(x, x') = \begin{cases} A y_L(x) y_R(x'), & x < x', \\ A y_L(x') y_R(x), & x > x', \end{cases}$$

with one shared amplitude A , because G must be continuous at x' : a jump in G would put a δ' into G'' , which the equation does not contain. Integrating $-(pG')' - qG = \delta$ over $[x' - \varepsilon, x' + \varepsilon]$ and letting $\varepsilon \rightarrow 0$ leaves the derivative jump

$$G'(x'^+, x') - G'(x'^-, x') = -\frac{1}{p(x')}. \quad (4.3.1)$$

Evaluating the jump from the piecewise form,

$$A[y_L(x')y_R'(x') - y_L'(x')y_R(x')] = A W[y_L, y_R](x') = -\frac{1}{p(x')},$$

A unit point source creates a jump (cont.)

so $A = -1/(p(x')W(x'))$. Abel's identity (Chapter 3) makes pW constant, so the value does not depend on where W is evaluated. The result is

$$G(x, x') = -\frac{y_L(x_{<})y_R(x_{>})}{p(x')W[y_L, y_R](x')}, \quad x_{<} = \min(x, x'), \quad x_{>} = \max(x, x'), \quad (4.3.2)$$

and $u(x) = \int_a^b G(x, x')f(x') dx'$ solves $Lu = f$ with the same boundary conditions. The jump condition (4.3.1) is local: it also applies when the interval is a circle (periodic angular coordinate), where the two solutions are matched periodically instead of by endpoint conditions. That jump is exactly what fixes modal coefficients when a field is expanded in eigenfunctions in a wave course.

The same function written in eigenfunctions

Alternatively expand G in the operator's own eigenfunctions. Let $L\phi_n = \lambda_n w\phi_n$ with real eigenfunctions and weighted norm $N_n = \int_a^b w\phi_n^2 dx$. Write $G(x, x') = \sum_n c_n \phi_n(x)$; then

$$\delta(x - x') = LG = \sum_n c_n \lambda_n w(x) \phi_n(x).$$

Multiplying by $\phi_m(x)$ and integrating, the weighted orthogonality gives $c_m \lambda_m N_m = \phi_m(x')$, hence

$$G(x, x') = \sum_{n=1}^{\infty} \frac{\phi_n(x) \phi_n(x')}{\lambda_n N_n}. \quad (4.3.3)$$

(For complex modes the conjugated orthogonality replaces $\phi_n(x')$ by $\overline{\phi_n(x')}$.) Equating (4.3.3) with (4.3.2) is the link between a *modal series* and the *two-solution* form of the same Green function.

Conversely, the eigenfunction expansion of an arbitrary f , $f(x) = \sum_n c_n \phi_n(x)$ with $c_n N_n = \int_a^b w(x') \phi_n(x') f(x') dx'$, can be rewritten as

$$f(x) = \int_a^b w(x') f(x') \left[\sum_{n=1}^{\infty} \frac{\phi_n(x) \phi_n(x')}{N_n} \right] dx'.$$

The same function written in eigenfunctions (cont.)

Since this must equal $\int \delta(x - x') f(x') dx' = f(x)$ for every f , the bracket is the delta source in the weighted inner product:

$$\sum_{n=1}^{\infty} \frac{\phi_n(x) \phi_n(x')}{N_n} = \frac{\delta(x - x')}{w(x)}. \quad (4.3.4)$$

This *completeness relation* is the precise sense in which the eigenfunctions span the space. A modal field expansion in a later wave course is exactly this statement with the weight and basis chosen by the geometry.

Gram–Schmidt for P_0, P_1, P_2

(Optional: Chapter 6 develops the full Legendre family and its norm. The short calculation below only shows how the first three polynomials arise by orthogonalizing $1, x, x^2, \dots$)

Recipe. Let $u_0, u_1, u_2, \dots$ be linearly independent. Set $v_0 = u_0$ and, for $k \geq 1$, subtract from u_k its projection onto each vector already built:

$$v_k = u_k - \sum_{j=0}^{k-1} v_j \frac{(u_k, v_j)}{\|v_j\|^2}, \quad k \geq 1. \quad (4.3.5)$$

Subtracting each projection removes the component in that direction. The following three-degree example shows the calculation.

First three Legendre polynomials. Use

$$(f, g) = \int_{-1}^1 f(x)g(x) dx \quad (4.3.6)$$

and orthogonalize $u_0 = 1$, $u_1 = x$, $u_2 = x^2$, so that (4.3.5) applies verbatim with $k = 1$ and $k = 2$.

Degree 0.

$$v_0(x) = 1, \quad \|v_0\|^2 = \int_{-1}^1 1 dx = 2. \quad (4.3.7)$$

Gram–Schmidt for P_0, P_1, P_2 (cont.)

Degree 1.

$$(x, v_0) = \int_{-1}^1 x \, dx = \left[\frac{x^2}{2} \right]_{-1}^1 = 0,$$

so

$$v_1(x) = x - v_0 \frac{(x, v_0)}{\|v_0\|^2} = x, \quad \|v_1\|^2 = \int_{-1}^1 x^2 \, dx = \frac{2}{3}. \quad (4.3.8)$$

Degree 2.

$$(x^2, v_0) = \int_{-1}^1 x^2 \, dx = \frac{2}{3}, \quad \frac{(x^2, v_0)}{\|v_0\|^2} = \frac{2/3}{2} = \frac{1}{3},$$
$$(x^2, v_1) = \int_{-1}^1 x^3 \, dx = 0.$$

Hence

$$v_2(x) = x^2 - v_0 \cdot \frac{1}{3} - v_1 \cdot 0 = x^2 - \frac{1}{3}. \quad (4.3.9)$$

Normalize by $P_n(1) = 1$:

$$P_0 = v_0 = 1, \quad P_1 = v_1 = x,$$

Gram–Schmidt for P_0, P_1, P_2 (cont.)

and, since $v_2(1) = 2/3$,

$$P_2(x) = \frac{v_2(x)}{v_2(1)} = \frac{x^2 - 1/3}{2/3} = \frac{1}{2}(3x^2 - 1). \quad (4.3.10)$$

Higher P_n follow from the same projection (or, more efficiently, from the Legendre recurrence / Rodrigues formula in Chapter 6).

Chapter summary

- Functions behave as vectors under $(u, v) = \int_a^b u \bar{v} dx$, or its weighted version. Our convention is linear in the first slot: $(cu, v) = c(u, v)$, $(u, cv) = \bar{c}(u, v)$.
- Orthogonality gives coefficients by projection: $a_n = (f, u_n)/\|u_n\|^2$. For $u_n = \sin(n\pi x)$ on $(0, 1)$, $\|u_n\|^2 = 1/2$.
- Integration by parts identifies both the formal adjoint and the boundary term. The expression $p_0 u'' + p_1 u' + p_2 u$ with real coefficients is formally self-adjoint when $p_1 = p_0'$, giving $(pu')' + qu$.
- Boundary conditions must make $[p(u\bar{v}' - u'\bar{v})]_a^b$ vanish. Under these conditions and a positive weight, $(pu')' + qu = -\lambda wu$ has real eigenvalues and weighted-orthogonal eigenfunctions for distinct eigenvalues.
- Multiplying a general second-order equation by an integrating factor puts its derivative terms into $(Qu')'$, with $Q = \exp \int (p_1/p_0) dx$. The Legendre, Bessel, and Hermite weights are 1, s , and e^{-x^2} .
- For $L = -(pu')' - qu$, the Green function solves $LG = \delta(x - x')$: it is continuous at x' , its derivative jumps by $-1/p(x')$, and it can be written either as $-y_L(x_<)y_R(x_>)/(pW)$ (4.3.2) or as the modal series (4.3.3); the basis also satisfies the completeness relation (4.3.4).